Limit theorems

Last modified: 2026-10-15 00:40:57 (PDT)

The Central Limit Theorem

Remark. The sum of many independent random variables, none of which dominates the others, has a distribution that is approximately bell-shaped, whatever the distributions of the individual variables. Theorem 0.1 makes this precise.

Theorem 0.1 (Central Limit Theorem)**.** *Let $X_1, X_2, \ldots$ be IID random variables with mean μ and finite variance $\sigma^2 > 0$, and let $S_n \stackrel{def}{=} \sum_{i=1}^{n} X_i$. Then for every real number z:*

$$\lim_{n\to\infty} \Pr\left(\frac{S_n - n\mu}{\sigma\sqrt{n}} \le z\right) = \Phi(z)$$

where $\Phi(z) \stackrel{def}{=} \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} e^{-u^2/2}\, du$ is the CDF of the standard normal distribution $\mathrm{N}(0,1)$.

Remark. This version is the Lindeberg–Lévy CLT; its proof is beyond these notes' scope (Billingsley 1995, Theorem 27.1). Other versions relax the IID assumption, which is why the informal statement asks only that no summand dominate.

Corollary 0.1 (Mean and variance of a sum of IID random variables)**.** *Let $X_1, \ldots, X_n$ be IID random variables, each discrete or continuous, with mean μ and finite variance σ^2, and let $S_n \stackrel{def}{=} \sum_{i=1}^{n} X_i$. Then:*

$$\mathrm{E}[S_n] = n\mu$$
$$\mathrm{Var}(S_n) = n\sigma^2$$

Proof

Proof. For the mean, apply linearity of expectation once per added summand:

$$\begin{aligned} \mathrm{E}[S_n] &= \sum_{i=1}^{n} \mathrm{E}[X_i] && \text{(linearity of expectation, applied } n-1 \text{ times)} \\ &= n\mu && \text{(each } X_i \text{ has mean } \mu) \end{aligned}$$

For the variance, apply the variance of a linear combination with every $a_i = 1$. For $i \neq j$, X_i and X_j are independent (take $A_k = \mathbb{R}$ for every other k in the definition of independence), so $\operatorname{Cov}(X_i, X_j) = 0$ for independent summands; and $\operatorname{Cov}(X_i, X_i) = \operatorname{Var}(X_i)$ (covariance of a variable with itself):

$$\begin{aligned} \operatorname{Var}(S_n) &= \sum_{i=1}^{n} \sum_{j=1}^{n} \operatorname{Cov}(X_i, X_j) && \text{(variance of a linear combination, all } a_i = 1) \\ &= \sum_{i=1}^{n} \operatorname{Cov}(X_i, X_i) + \sum_{\substack{i,j=1 \\ i \neq j}}^{n} \operatorname{Cov}(X_i, X_j) && \text{(split off the terms with } i = j) \\ &= \sum_{i=1}^{n} \operatorname{Var}(X_i) + 0 && \text{(covariance with itself; independent summands)} \\ &= n\sigma^2 && \text{(each } X_i \text{ has variance } \sigma^2) \end{aligned}$$

□

Remark. In practice, Theorem 0.1 justifies approximating S_n by a normal distribution with mean $n\mu$ and variance $n\sigma^2$, which by Corollary 0.1 are exactly the mean and variance of S_n.

Example 0.1 (The sum of five dice)**.** A single fair die roll has the discrete uniform distribution on $\{1, \dots, 6\}$, which is flat, not bell-shaped (Figure 1). Its mean is $\mu = 3.5$, and its variance is

$$\begin{aligned} \sigma^2 &= \sum_{x=1}^{6} (x - 3.5)^2 / 6 \\ &= 35/12. \end{aligned}$$

```r
dice_sum_pmf <- function(n_dice) {
  totals <- rowSums(expand.grid(rep(list(1:6), n_dice)))
  probs <- prop.table(table(totals))
  tibble::tibble(total = as.numeric(names(probs)), p = as.vector(probs))
}

dice_plot <-
  ggplot2::ggplot(mapping = ggplot2::aes(x = total, y = p)) +
  ggplot2::xlab("sum of dice (x)") +
  ggplot2::ylab("Probability of outcome, Pr(X=x)") +
  ggplot2::expand_limits(y = 0)

dice_plot + ggplot2::geom_col(data = dice_sum_pmf(1))
```

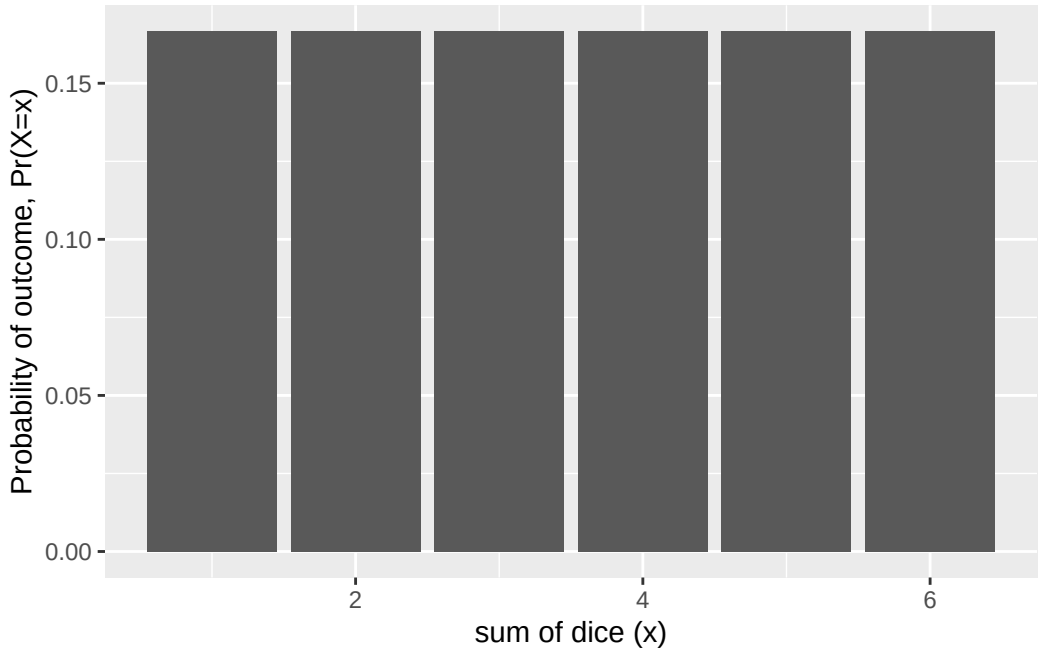


Figure 1: Distribution of the outcome of one die

The sum of five independent rolls, S_5, is already close to bell-shaped (Figure 2).

```r
pmf_five_dice <- dice_sum_pmf(5)
dice_plot + ggplot2::geom_col(data = pmf_five_dice)
```

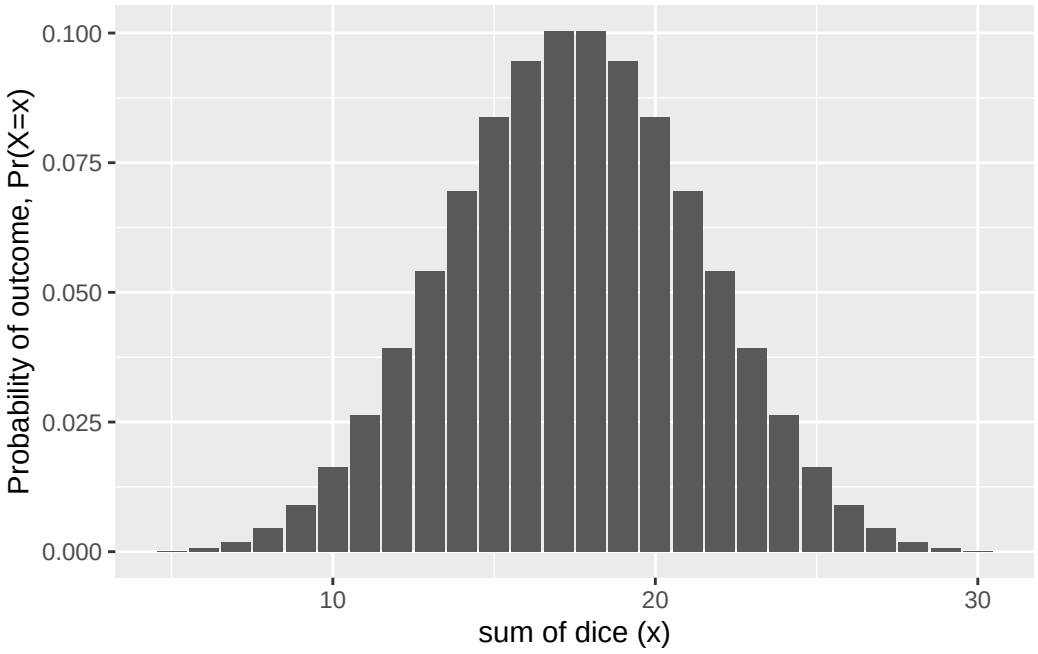


Figure 2: Distribution of the sum of five dice

For example, the exact probability that five dice total at most 15 is $\Pr(S_5 \le 15) = 0.3052$. The normal approximation that Theorem 0.1 suggests, with mean $5 \cdot 3.5 = 17.5$ and variance $5 \cdot 35/12 \approx 14.58$, evaluated at 15.5 to account for S_5 taking only integer values, gives $\Phi\big((15.5 - 17.5)/\sqrt{14.58}\big) \approx 0.3002$.

Example 0.2 (Sample means of car horsepower)**.** The Auto data record the horsepower and fuel economy of 392 car models sold between 1970 and 1982. They come from the StatLib library at Carnegie Mellon University and ship with the book *An Introduction to Statistical Learning*, in the Python package `ISLP` (`ISLP.load_data("Auto")`) and the R package `ISLR2` (`ISLR2::Auto`). This site has a copy at `data/auto.csv`, and both code tabs below read that file.

Horsepower is not bell-shaped: a few powerful cars give it a long right tail. To apply Theorem 0.1, treat each draw of one car at random as one IID observation X_i, so that μ and σ^2 are the mean and variance of the 392 horsepower values. The sum S_n has mean $n\mu$ and variance $n\sigma^2$ by Corollary 0.1, so the sample mean S_n/n has mean μ and standard deviation $\sigma/\sqrt{n}$. The code below draws 10,000 random samples of each size $n \in \{1, 5, 30\}$, with replacement, and computes each sample's mean.

Python

```python
import numpy as np
import pandas as pd

hp = pd.read_csv("data/auto.csv")["horsepower"].to_numpy(dtype=float)
mu, sigma = hp.mean(), hp.std(ddof=0)

rng = np.random.default_rng(571)
sizes = [1, 5, 30]
sample_means = {n: rng.choice(hp, size=(10_000, n)).mean(axis=1) for n in sizes}

def skewness(x):
    return np.mean(((x - x.mean()) / x.std()) ** 3)

pd.DataFrame({
    "n": sizes,
    "sd of sample means": [sample_means[n].std(ddof=1) for n in sizes],
    "sigma / sqrt(n)": [sigma / np.sqrt(n) for n in sizes],
    "skewness of sample means": [skewness(sample_means[n]) for n in sizes],
})
#>     n  sd of sample means  sigma / sqrt(n)  skewness of sample means
#> 0   1           38.691521        38.442033                  1.080767
#> 1   5           17.265666        17.191800                  0.522243
#> 2  30            6.971269         7.018523                  0.164940
```

R

```r
hp <- read.csv("data/auto.csv")$horsepower
mu <- mean(hp)
sigma <- sqrt(mean((hp - mu)^2))

set.seed(571)
sizes <- c(1, 5, 30)
sample_means <- lapply(sizes, function(n) {
  replicate(10000, mean(sample(hp, n, replace = TRUE)))
})

skewness <- function(x) mean(((x - mean(x)) / sqrt(mean((x - mean(x))^2)))^3)

data.frame(
  n = sizes,
  sd_of_sample_means = sapply(sample_means, sd),
  sigma_over_root_n = sigma / sqrt(sizes),
  skewness_of_sample_means = sapply(sample_means, skewness)
)
#>    n sd_of_sample_means sigma_over_root_n skewness_of_sample_means
#> 1  1          38.611630         38.442033                1.0696654
#> 2  5          16.987684         17.191800                0.4804001
#> 3 30           7.057655          7.018523                0.1991387
```

Figure 3 shows a histogram of the 10,000 sample means for each n, with the normal density $\mathrm{N}(\mu, \sigma^2/n)$ from Theorem 0.1 drawn over it.

```r
n_labels <- paste("n =", sizes)
means_long <- tibble::tibble(
  n = factor(
    rep(n_labels, each = length(sample_means[[1]])),
    levels = n_labels
  ),
  sample_mean = unlist(sample_means)
)
normal_curves <- seq_along(sizes) |>
  lapply(function(k) {
    se <- sigma / sqrt(sizes[k])
    grid <- seq(mu - 4 * se, mu + 4 * se, length.out = 200)
    tibble::tibble(
      n = factor(n_labels[k], levels = n_labels),
      sample_mean = grid,
      density = dnorm(grid, mu, se)
    )
  }) |>
  dplyr::bind_rows()

ggplot2::ggplot(means_long, ggplot2::aes(x = sample_mean)) +
  ggplot2::geom_histogram(
    ggplot2::aes(y = ggplot2::after_stat(density)),
    bins = 40,
    fill = "grey70",
    color = "white"
  ) +
  ggplot2::geom_line(
    ggplot2::aes(y = density),
    data = normal_curves,
    linewidth = 1
  ) +
  ggplot2::facet_wrap(ggplot2::vars(n), scales = "free") +
  ggplot2::labs(x = "sample mean of horsepower", y = "density") +
  ggplot2::theme_minimal()
```

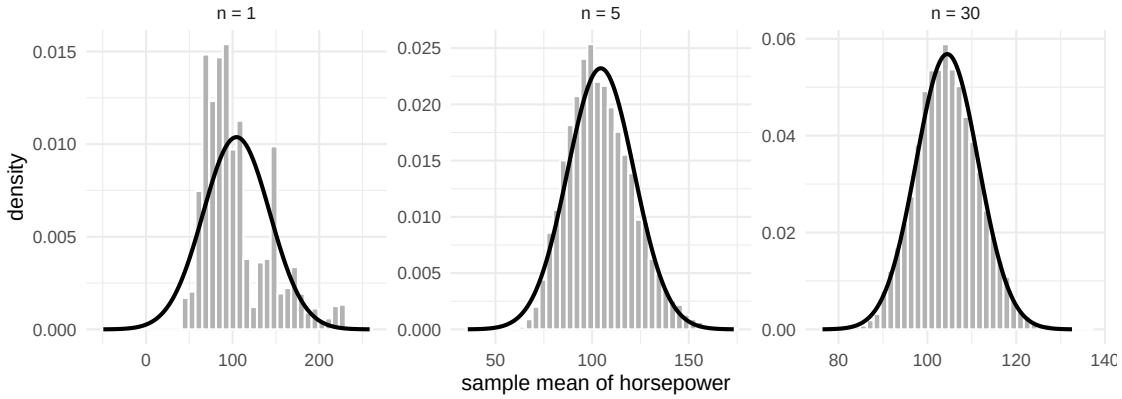


Figure 3: Histograms of 10,000 sample means of horsepower, for samples of n = 1, 5 and 30 cars drawn with replacement from the Auto data. The curve is the normal density with mean mu and standard deviation sigma over the square root of n.

The sample means for $n = 1$ are just single cars, so they are skewed. By $n = 30$ the histogram is close to the normal curve. The standard deviation of the sample means is within 1.2% of $\sigma/\sqrt{n}$ at every n, as Corollary 0.1 predicts. The skewness of the sample means, which is zero for a normal distribution, falls from 1.07 at $n = 1$ to 0.48 at $n = 5$ and 0.2 at $n = 30$.

i Watch

5MinuteAI explains limit theorems and gives a proof of the CLT, following the University of Washington's CSE 312 (5MinuteAI, n.d.).

Tail bounds

Theorem 0.2 (Markov's inequality)**.** *Let X be a random variable with $X \geq 0$ and $\mathrm{E}[X] < \infty$. Then for every $a > 0$:*

$$\Pr(X \geq a) \leq \frac{\mathrm{E}[X]}{a}$$

i Proof

Proof. Because $X \geq 0$, every outcome satisfies $X \geq a \cdot \mathbb{1}_{\{X \geq a\}}$: when $X \geq a$ the right side is $a \leq X$, and otherwise it is $0 \leq X$. The expectation of a nonnegative random variable is

nonnegative, so by linearity of expectation, and because the expectation of an indicator is the probability of its event,

$$\mathrm{E}[X] - a \cdot \Pr(X \geq a) = \mathrm{E}\left[X - a \cdot \mathbb{1}_{\{X \geq a\}}\right] \geq 0$$

and dividing by $a > 0$ gives the claim. □

Remark. The inequality carries Markov's name by convention. Chebyshev's inequality, which follows from it, is older (Bienaymé 1853; Tchébychef 1867); see (Seneta 2013) for the history of the attribution. A modern treatment is in (Blitzstein and Hwang 2019).

Theorem 0.3 (Chebyshev's inequality)**.** *Let X be a random variable with mean μ and finite variance $\sigma^2 > 0$. Then for every $k > 0$:*

$$\Pr(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$$

Proof

Proof. The random variable $(X - \mu)^2$ is nonnegative, and its expectation is σ^2. Because $|X - \mu| \geq k\sigma$ exactly when $(X - \mu)^2 \geq k^2\sigma^2$, Theorem 0.2 gives

$$\Pr(|X - \mu| \geq k\sigma) = \Pr((X - \mu)^2 \geq k^2\sigma^2) \leq \frac{\sigma^2}{k^2\sigma^2} = \frac{1}{k^2}$$

□

Remark. Theorem 0.3 needs only a mean and a variance, not a distribution, so it holds for every distribution that has them. That generality makes it loose: for a normal distribution, the chance of landing more than two standard deviations from the mean is about 5%, but the bound at $k = 2$ is 25%.

Remark. Bienaymé derived this bound in 1853 (Bienaymé 1853), and Chebyshev proved a more general form in 1867 (Tchébychef 1867). A modern treatment is in (Blitzstein and Hwang 2019).

Example 0.3 (How loose is Chebyshev's bound for car horsepower?)**.** Take the 392 horsepower values in the Auto data described in Example 0.2 as the whole population, and draw one car at random. For each k, the code below counts the share of cars whose horsepower is at least k standard deviations from the mean, and sets it beside the bound

$1/k^2$ from Theorem 0.3.

```
hp <- read.csv("data/auto.csv")$horsepower
mu <- mean(hp)
sigma <- sqrt(mean((hp - mu)^2))

k <- c(1.5, 2, 3)
share_beyond <- sapply(k, function(kk) mean(abs(hp - mu) >= kk * sigma))
data.frame(k = k, share_of_cars = share_beyond, chebyshev_bound = 1 / k^2)
#>     k share_of_cars chebyshev_bound
#> 1 1.5    0.09948980       0.4444444
#> 2 2.0    0.04336735       0.2500000
#> 3 3.0    0.01275510       0.1111111
```

The bound holds at every k, as Theorem 0.3 guarantees. At $k = 2$ the share of cars is 4.3%, against a bound of 25%, and at $k = 3$ it is 1.3% against 11.1%. The bound is loose because it must hold for every distribution with this mean and variance.

Exercise 0.1 (How many cars to estimate average fuel economy)**.** Let $X_1, \ldots, X_n$ be IID draws of the fuel economy (`mpg`) of a car chosen at random from the Auto data, with mean μ and variance σ^2, and let $\bar{X}_n \stackrel{\text{def}}{=} \frac{1}{n}\sum_{i=1}^{n} X_i$.

a. Use Corollary 0.1 and Theorem 0.3 to show that for every $\epsilon > 0$,

$$\Pr\left(\left|\bar{X}_n - \mu\right| \geq \epsilon\right) \leq \frac{\sigma^2}{n\epsilon^2}$$

b. Compute σ^2 for the `mpg` column of `data/auto.csv`, treating the 392 cars as the whole population. Find the smallest n for which the bound in (a) guarantees $\Pr\left(\left|\bar{X}_n - \mu\right| \geq 1\right) \leq 0.05$.

c. Simulate 10,000 samples of that size drawn with replacement, and count how many sample means miss μ by 1 or more. Why is the count so far below the 5% the bound allows?

Solution

Solution 0.1.

a. By Corollary 0.1, $\mathrm{Var}(S_n) = n\sigma^2$ for $S_n \stackrel{\text{def}}{=} \sum_{i=1}^{n} X_i$, so $\bar{X}_n = S_n/n$ has mean μ and variance σ^2/n (a constant factor c scales a variance by c^2). Apply Theorem 0.3 to $\bar{X}_n$ with $k\sigma/\sqrt{n} = \epsilon$, that is, $k = \epsilon\sqrt{n}/\sigma$, so $1/k^2 = \sigma^2/(n\epsilon^2)$.

b. Setting $\sigma^2/(n \cdot 1^2) \leq 0.05$ gives $n \geq 20\sigma^2$.

c. See the code below.

```
mpg <- read.csv("data/auto.csv")$mpg
mu <- mean(mpg)
sigma2 <- mean((mpg - mu)^2)
n_min <- ceiling(20 * sigma2)

set.seed(571)
xbar <- replicate(10000, mean(sample(mpg, n_min, replace = TRUE)))
misses <- sum(abs(xbar - mu) >= 1)
normal_approx <- 2 * pnorm(-1 / sqrt(sigma2 / n_min))
c(variance = sigma2, n = n_min, misses = misses, normal_approx = normal_approx)
#>      variance             n        misses normal_approx
#>  6.076274e+01  1.216000e+03  0.000000e+00  7.694705e-06
```

The population variance of `mpg` is 60.8, so $n = 1216$ cars suffice. Out of 10,000 samples of that size, 0 sample means missed μ by 1 or more, against the 5% (500 samples) the bound allows. The bound is loose because the sample mean of this many cars is nearly normal (Theorem 0.1), and the normal approximation puts the chance of a miss at about 7.7×10^{-6}; Chebyshev's inequality must hold for every distribution, as in Example 0.3.

The law of large numbers

Definition 0.1 (Convergence in probability)**.** A sequence of random variables $Y_1, Y_2, \ldots$ **converges in probability** to a constant c if, for every distance $\varepsilon > 0$, the probability that Y_n lies at least ε away from c goes to zero as n grows:

$$\lim_{n \to \infty} \Pr(|Y_n - c| \geq \varepsilon) = 0$$

We write $Y_n \xrightarrow{p} c$.

Source

Casella and Berger (2002, Def. 5.5.1).

Theorem 0.4 (Weak law of large numbers)**.** *Let $X_1, X_2, \ldots$ be IID random variables with mean μ and finite variance $\sigma^2 > 0$, and let $\bar{X}_n \overset{def}{=} \frac{1}{n} \sum_{i=1}^{n} X_i$ be the sample mean. Then the sample mean converges in probability to μ:*

$$\bar{X}_n \xrightarrow{p} \mu$$

Source

Casella and Berger (2002, Thm. 5.5.2).

Proof

Proof. By Corollary 0.1, $\mathrm{E}[S_n] = n\mu$ and $\mathrm{Var}(S_n) = n\sigma^2$, where $S_n \overset{\text{def}}{=} \sum_{i=1}^{n} X_i$, so $\bar{X}_n = S_n/n$ has

$$\begin{aligned}
\mathrm{E}\big[\bar{X}_n\big] &= \frac{1}{n}\,\mathrm{E}[S_n] && \text{(linearity of expectation)} \\
&= \frac{1}{n} \cdot n\mu && \text{(mean of a sum of IID variables)} \\
&= \mu && \text{(cancel } n\text{)} \\
\mathrm{Var}\big(\bar{X}_n\big) &= \frac{1}{n^2}\,\mathrm{Var}(S_n) && \text{(variance of a constant multiple)} \\
&= \frac{1}{n^2} \cdot n\sigma^2 && \text{(variance of a sum of IID variables)} \\
&= \frac{\sigma^2}{n} && \text{(cancel one factor of } n\text{)}
\end{aligned}$$

Fix $\varepsilon > 0$ and let $k \overset{\text{def}}{=} \varepsilon\sqrt{n}/\sigma$, so that $k\sigma/\sqrt{n} = \varepsilon$. Chebyshev's inequality for $\bar{X}_n$, whose standard deviation is $\sigma/\sqrt{n}$, gives

$$\begin{aligned}
0 \le \Pr\big(\big|\bar{X}_n - \mu\big| \ge \varepsilon\big) &= \Pr\left(\big|\bar{X}_n - \mu\big| \ge k\frac{\sigma}{\sqrt{n}}\right) && \text{(definition of } k\text{)} \\
&\le \frac{1}{k^2} && \text{(Chebyshev's inequality)} \\
&= \frac{\sigma^2}{n\varepsilon^2} && \text{(substitute } k\text{)}
\end{aligned}$$

The bound goes to zero as $n \to \infty$, so the probability, squeezed between zero and the bound, goes to zero. This holds for every $\varepsilon > 0$. □

References

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